3.3.2 X-ray reflectometry (XRR)

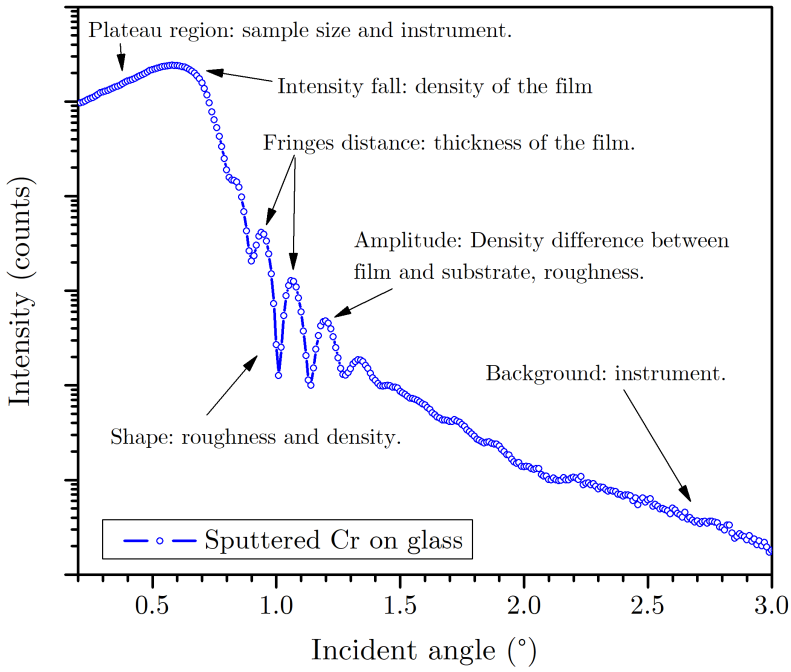

Figure 3.14: XRR curve of a 55 nm sputtered Cr film on glass. The plot illustrates key structural features provided by XRR data.

X-ray reflectivity (XRR) is a non-destructive technique that involves irradiating the surface of a sample with a monochromatic X-ray beam of known wavelength at grazing incidence angles. The reflected X-ray intensity is collected as a function of the incident angle γ and is used to determine thin-film properties including thickness, density, and surface roughness. This technique can also be used to analyze multilayer film architectures, including individual layer properties and interface roughness.

Unlike in the optical regime, where the refractive index of materials is typically greater than one, the refractive index n of a solid for X-ray radiation is slightly less than one. As a result, a critical angle θ_C for total external reflection is reached at very small incidence angles. Depending on the X-ray wavelength and energy, θ_C typically falls within the grazing incidence range of 0° to 2°.

The analysis of the reflected intensity curves as a function of incident angle between 0° and 6° reveals different features for values below, equal to, and greater than the critical angle θ_C of the medium. For incident angles below θ_C, the X-ray radiation is totally reflected. When the incident angle is equal to θ_C, the radiation is partially reflected and propagates along the surface of the sample. As the incident angle increases above θ_C, the intensity of the X-ray radiation decreases rapidly, proportional to a factor of θ^4 [25]. In this region, the incident X-ray radiation

penetrates the film and is reflected at the interface between the film and the substrate. The multiple reflections generate an interference pattern in which:

- the periodicity of the fringes is proportional to the film thickness,
- the intensity is proportional to the surface/interface roughness,
- the amplitude is proportional to the density contrast between film and substrate.

X-ray reflectivity is a sensitive technique that uses this interference to extract structural information from complex multilayered systems. Reflections occur at all internal interfaces, and even in simple bilayers, intricate interference patterns can emerge. By comparing experimental data with theoretical models, the density, thickness, and roughness of each layer can be quantitatively determined [25].

Geometrical correction: The "footprint effect"

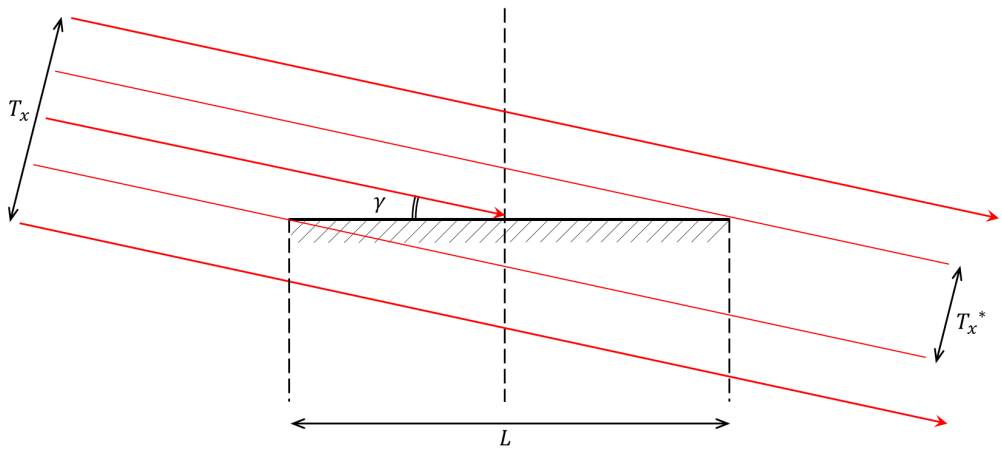

Figure 3.15: Geometric schematic of the footprint effect, i.e., a reduction in reflected intensity in XRR measurements at small incident angles, caused by only a fraction of the incoming beam irradiating the sample surface. A fraction T_x^* of the total beam thickness irradiates the entire sample.

The reflected X-ray intensity is given by:

$$I(\gamma) = R(\gamma)\, I_0 \tag{3.24}$$

where $I(\gamma)$ is the measured reflected X-ray intensity, $R(\gamma)$ is the reflection coefficient, and I_0 is the incident beam intensity.

The shape of the XRR curves in the region of total external reflection ideally appears as a plateau. However, a broad feature may arise in this region due to geometrical limitations when the beam footprint exceeds the sample size. The width of the beam footprint, i.e., the area irradiated on the sample by the X-ray beam, and the corrected X-ray intensity $I(\gamma)$ would ideally be given by:

$$F_b = \frac{T_x}{\sin\gamma} \tag{3.25}$$

$$I(\gamma) = R(\gamma)\, I_0\, \frac{T_x^*}{T_x} = R(\gamma)\, I_0\, \frac{L \sin\gamma}{T_x} \tag{3.26}$$

Where γ is the incident angle, T_x is the beam thickness determined by the aperture of the front slits [26], and T_x^* is the effective portion of the beam irradiating the sample. A portion of the beam is lost, leading to a decrease in the reflected X-ray intensity $I(\gamma)$ by a factor of T_x^*/T_x. This phenomenon is commonly referred to as the *footprint effect*.

Equation 3.26 is valid only when the centers of the beam and sample are aligned. However, this may not always be the case, and lateral displacement must be considered. Figure 3.16 illustrates the three scenarios that arise when the center of the X-ray beam is translated by a distance Δ relative to the center of the sample. These cases are defined by two critical angles, γ_1 and γ_1, at which the edges of the beam and the sample match:

$$\sin\gamma_1 = \frac{T_x}{L + 2\Delta} \quad ; \quad \sin\gamma_2 = \frac{T_x}{L - 2\Delta} \tag{3.27}$$

The reflected X-ray intensity requires correction depending on the angular regime, defined by the value of γ relative to the critical angles γ_1 and γ_2. The corresponding expressions are:

$$I(\gamma) = \begin{cases} R(\gamma)\, I_0\, \dfrac{L}{T_x} \sin\gamma & \text{if } \sin\gamma < \dfrac{T_x}{L + 2\Delta} \\ R(\gamma)\, I_0 \left[\dfrac{1}{2} + \left(\dfrac{L - 2\Delta}{2T_x}\right)\right] \sin\gamma & \text{if } \dfrac{T_x}{L + 2\Delta} < \sin\gamma < \dfrac{T_x}{L - 2\Delta} \\ R(\gamma)\, I_0 & \text{if } \sin\gamma > \dfrac{T_x}{L - 2\Delta} \end{cases} \tag{3.28}$$

Although a detailed analysis of effective beam thickness and sample alignment effects on XRR data is beyond the scope of this thesis, the preceding discussion shows the importance of applying geometrical correction in the plateau region. Without this correction, the loss of reflected intensity at low incident angles γ may difficult proper normalization. This, in turn, affects the weighting of data near the critical angle θ_C, which may be essential for accurate determination of film density.

XRR measurement conditions

The XRR measurements were performed using a PANalytical X'Pert PRO MRD system with a Cu Kα X-ray beam of wavelength $\lambda = 1.54$ Å and a beam cross-section of 100 μm. The specular XRR intensities were collected over an angular 2θ range from 0° to 6°. The samples were cut into squares of 25 mm × 25 mm. All XRR data were processed using the PANalytical X'Pert Reflectivity software (Malvern PANalytical B.V., Netherlands).

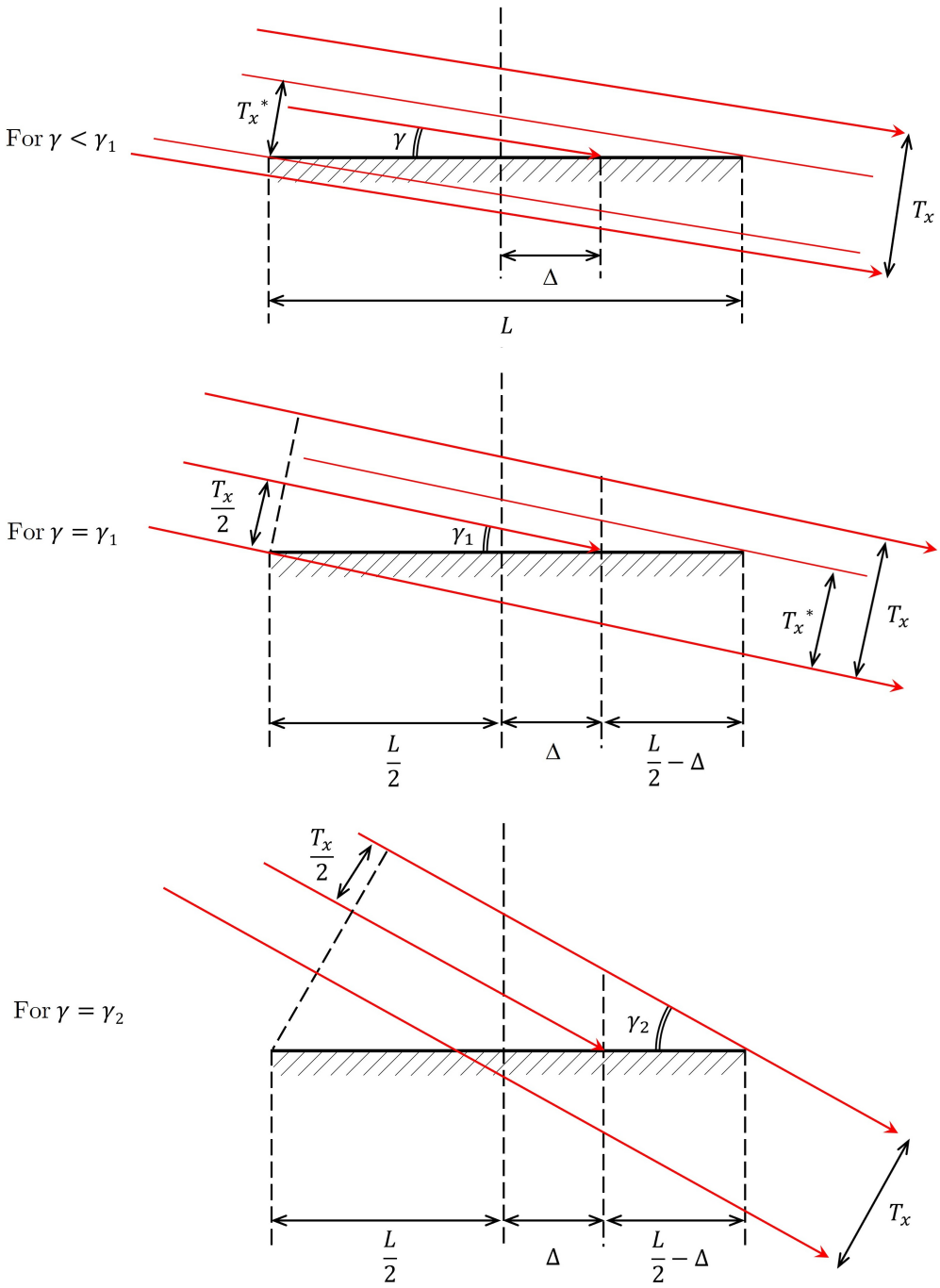

Figure 3.16: Geometric schematics of the footprint effect in XRR measurements as function of angle of incidence when the center of the beam is translated a value Δ from the center of the sample. The angles γ_1 and γ_2 are the angles in which the boundaries of the beam and sample match.

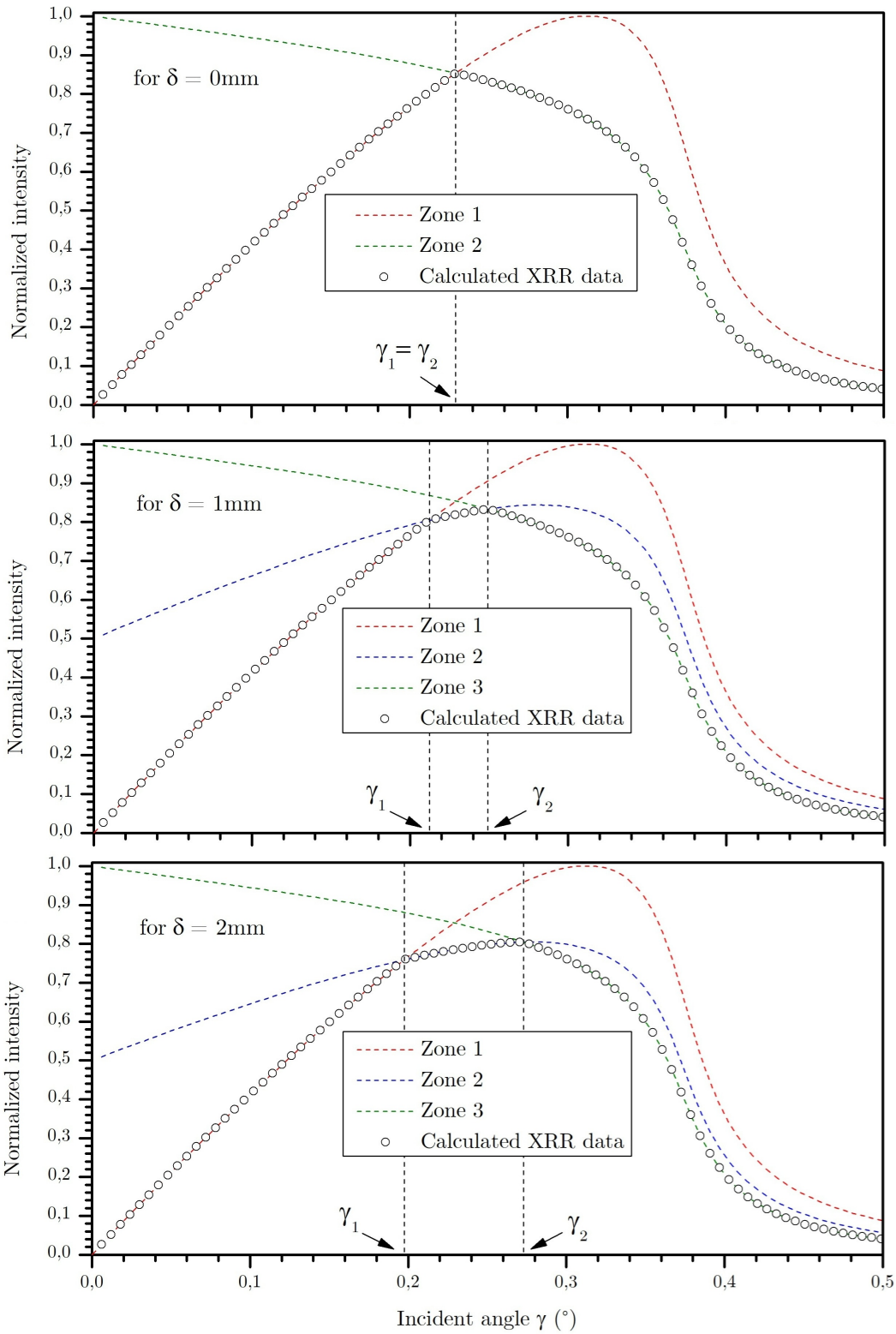

Figure 3.17: Calculated XRR intensity curves for a Cr film deposited on a 25 mm × 25 mm glass substrate, showing the effect of beam–sample misalignment (Δ = 0, 1, and 2 mm). A beam width of 0.1 mm was used. The results show the importance of applying geometrical correction at low incident angles γ, where the footprint effect reduces the reflected intensity. Zone transitions correspond to the angular regimes defined by the critical angles γ_1 and γ_2.